Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/18/2024
PI Name:	Stray, Stephen J. Ph.D.		
Project Title:	Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Jackson	**State:**	MS
Zip Code:	39216	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	01/01/2020	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	The previous Principal Investigator (PI) was Dr. Ritesh Tandon at the University of Mississippi Medical Center from January 2020-October 2023. Dr. Stephen J. Stray became PI on Oct 5, 2023.		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Mehta, Satish Ph.D. (NASA Johnson Space Center) Simpson, Richard Ph.D. (University of Arizona)		
Grant/Contract No.:	80NSSC19K1603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE: This project (grant number 80NSSC19K1603), is a continuation of the investigation titled, "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication", with Dr. Stephen J. Stray from the University of Mississippi Medical Center as the new Principal Investigator (PI).
Opportunistic pathogens such as latent viruses pose a unique hazard to crew members because of the ineffectiveness of preflight quarantine periods and decreased immune function. Latent herpesviruses reactivate and shed in higher quantities during and after long and short spaceflights. Human cytomegalovirus (CMV) is an important herpesvirus from a public health viewpoint that causes severe systemic infections upon primary infection or on reactivation from latency. The few antiviral drugs available to treat CMV infection suffer from safety, efficacy, and resistance issues, and no vaccine against CMV is currently available. As NASA and collaborating space agencies plan for future exploration class missions (i.e., to Mars or an asteroid), eliminating the risk of an adverse health event due to alterations in virus latency, reactivation, and lytic replication is paramount. Evidence of impaired control of latent viral infections has been documented during short (i.e., 14-day Space Shuttle missions) as well as long (i.e., 6-month International Space Station missions) duration spaceflight missions, with the magnitude of this response being positively associated with time in flight. Shedding of herpesviruses including Epstein–Barr virus (EBV), varicella-zoster virus (VZV), and CMV was detected in >30% of astronauts in a recent study (Mehta et al., 2017) looking at long-duration spaceflight (60–180 days). These viruses reactivated independently of each other and reactivation frequency increased in duration, and amplitude (viral copy numbers) when compared to short duration (10 to 16 days) space shuttle missions. Thus, it is easy to speculate that the substantially longer durations being proposed for exploration class missions (i.e., up to 3 years) will further increase the risk of an adverse health event due to increased virus reactivation and/or replication. Add increased radiation exposure (both duration and magnitude), limited clinical care, and the impossibility of rapidly returning a casualty back to Earth to the list of potential hazards and it is clear to see why obtaining a better understanding of virus replication under space-like conditions is critical to successfully and safely send humans to, and return them from, these remote extraterrestrial destinations. In this proposal, we aim to determine the impact of space radiation on CMV latency, reactivation, and lytic replication. While most studies on this topic in the past have focused on changes in the human immune system that could induce virus reactivation, we will approach this topic from both virus and host standpoints. We will study differences in CMV latency and reactivation before and after radiation exposure and determine if these changes can be correlated with specific changes in virus genome and/or transcriptome. Also, because virus control, dissemination, and resolution are ultimately governed by the competency, or lack thereof, of the host immune system, we will determine the impact of pre-exposing NK-cells and T-cells to space radiation on their ability to control active CMV infections. In this project, we will focus on a single variable: space radiation, rather than looking at multiple variables such as microgravity, physical and psychological stresses of space travel, diet, medications, or other factors associated with spaceflight. The overarching hypothesis is that modeled space radiation will induce specific changes in CMV genome/transcriptome or immune control of infection that will induce CMV reactivation and/or increase in virus replication and pathogenicity. The specific aims to test this hypothesis are: Specific Aim 1: Determine the impact of space radiation on CMV genome and transcriptome.

Specific Aim 2: Correlate radiation induced changes in CMV genome and/or transcriptome with CMV latency, reactivation, and lytic replication.

Specific Aim 3: Determine the impact of space radiation on immune cell control in a mouse model of CMV infection.

Reference: Mehta SK, Laudenslager ML, Stowe RP, Crucian BE, Feiveson AH, Sams CF, Pierson DL. (2017) Latent virus reactivation in astronauts on the international space station, NPJ Microgravity 3, 11.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

December 2024 Update - Compiled from the technical report submitted to NASA [Ed., 12/18/24.]
Work on this project was severely delayed due to the COVID pandemic. Dr Tandon left the University of Mississippi Medical Center (UMMC) in 2023 to become a Program Officer at the National Institute of Health (NIH) and has no further involvement with the project. This report was prepared by Stephen Stray, Ph.D., who replaced Dr Tandon as Principal Investigator (PI), assisted by Co-Investigator Dr. Satish Mehta. [Ed. Note: For prior reporting, please see "Effect of Space Radiation on Cytomegalovirus Reactivation and Lytic Replication" under Dr. Tandon: https://].

Specific Aim 1

Irradiation studies and analysis of copy number were completed satisfactorily (Mehta). Cytomegalovirus (CMV) genomic DNA was sequenced from a number of conditions, but no evidence of mutation was found. Plaque assays were initiated, but not completed prior to funds running out due to illness of a staff member (Tandon).

Specific Aim 2

Transcriptome analysis was conducted on irradiated cultures, which showed the upregulation of transcription of the UL49 ORF. This finding suggests that this ORF may have an important role in the switch from latent to lytic replication. UL49 may be a direct mediator of exit from latency and the re-establishment of lytic replication; it may be regulated by the “master switch” controlling latency and the re-establishment of lytic replication, or both. To test the hypothesis that UL49 plays a key role in the exit from latency and re-establishment of lytic infection, we wished to create a UL49 knock-out in TB40E, then test the ability of this knockout to undergo both lytic replication in human foreskin fibroblasts (HFF) and to establish latency and be reactivated in the Kasumi cell model. Mutations deleting UL49 in the TB40 BAC genomic clone were made and verified by DNA sequencing, but no viable virus could be rescued. This finding strongly suggests that UL49 deletion is lethal in the HFF model. Attempts to rescue the UL49-deleted TB40E mutant by expressing UL49 in trans were not successful, as we could never get UL49 to express. These technical problems in constructing the TB40E knockout were not resolved before the exhaustion of the grant funding.

Specific Aim 3

Arrangements were made to perform mouse irradiation experiments described in the grant application to determine the effect of radiation on immune control of murine CMV. No results are available at this time.

Bibliography Type:	Description: (Last Updated: 12/18/2024)
Abstracts for Journals and Proceedings	Mehta S, Bustos-Lopez S, Nelman-Gonzalez M, Rooney B, Xi C, Plante I, Tandon R, Crucian B. "Effect of simulated cosmic radiation on cytomegalovirus reactivation and lytic replication." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Articles in Peer-reviewed Journals	Mehta SK, Diak DM, Bustos-Lopez S, Nelman-Gonzalez M, Chen X, Plante I, Stray SJ, Tandon R, Crucian BE. "Effect of simulated cosmic radiation on cytomegalovirus reactivation and lytic replication." Int J Mol Sci. 2024 Sep 26;25(19):10337. https://doi.org/10.3390/ijms251910337 ; PMID: 39408667; PMCID: PMC11477029 , Sep-2025